Operating System Question Bank: Unit 1

Here are the answers formatted as per your request:

Level A: Easy Questions (2 Marks Each)

Q1. What is an Operating System (OS)?

An Operating System (OS) is system software that acts as an interface between computer hardware and the user. It manages hardware resources and provides services for application software.

Q2. List two primary functions of an Operating System.

1. Resource Management: Manages CPU, memory, and I/O devices.
2. User Interface: Provides an interface for users to interact with the system.

Q3. What is the difference between system software and application software?

- System Software: Facilitates the functioning of hardware and basic tasks (e.g., Operating System).
- Application Software: Performs specific tasks for users (e.g., MS Word).

Q4. Name two resources managed by an Operating System.

1. CPU (Processor): Allocates processing time.
2. Memory: Manages storage and retrieval of data.

Q5. Define the term "process" in the context of an Operating System.

A process is a program in execution, including its code, data, and resources required for execution.

Q6. What is a batch operating system?

A batch operating system executes jobs in batches without user interaction during processing.

Q7. Explain what a time-sharing operating system is.

A time-sharing OS allows multiple users to use the system simultaneously by sharing CPU time in small intervals.

Q8. What are real-time operating systems used for?

Real-time OSs are used in applications requiring immediate processing, such as medical devices and automotive systems.

Q9. What is a multiprocessor system?

A multiprocessor system has two or more CPUs that work collaboratively to enhance performance and reliability

Q10. What does it mean for an operating system to be multiuser?

A multiuser OS allows multiple users to access and use the computer resources concurrently.

Q11. What is a multithreaded system?

A multithreaded system allows a process to contain multiple threads, each performing a different task simultaneously.

Q12. How does a multiprocessor system improve performance?

By parallelizing tasks, multiprocessor systems enhance speed and reliability while handling more processes concurrently.

Q13. What is a layered structure in an operating system?

A layered structure organizes an OS into layers, each with a specific function, simplifying development and debugging.

Q14. List two advantages of a layered operating system structure.

1. Modularity: Easier to develop and maintain.
2. Isolation: Errors in one layer do not affect others.

Q15. What is the purpose of system components in an OS?

System components provide the core functionality for resource management, process control, and user interface.

Q16. Name two common system components in an OS.

1. Process Management: Handles processes.
2. Memory Management: Manages memory allocation.

Q17. What is a kernel in an operating system?

The kernel is the core component of the OS, managing hardware, processes, memory, and system calls.

Q18. What is a monolithic kernel?

A monolithic kernel integrates all OS functions like process management, file management, and device drivers in one code base.

Q19. Give one advantage and one disadvantage of a monolithic kernel.

- Advantage: High performance.
- Disadvantage: Lack of modularity and security risks.

Q20. What is a microkernel?

A microkernel is a minimal kernel that performs essential OS functions like IPC and basic process management, delegating other services to user space.

Level B: Intermediate Questions (5 Marks Each)

Q21. Explain the role of an Operating System in managing hardware and software resources. Provide examples.

The OS ensures efficient use of system resources by:

1. Hardware Resource Management:
   - Allocates CPU time to processes using scheduling algorithms (e.g., Priority Scheduling).
   - Manages memory allocation, ensuring each process has sufficient space and preventing conflicts.
   - Controls I/O devices, facilitating communication between hardware and software.
2. Software Resource Management:
   - Coordinates application execution through system calls.
   - Handles process synchronization to avoid deadlocks.

Example: It enables multitasking, such as running a video player and a web browser concurrently.

Q22. Discuss the difference between user mode and kernel mode in an Operating System. Why is this distinction important?

1. User Mode:
   - Restricted access for running applications.
   - Prevents direct hardware access to ensure safety.
2. Kernel Mode:
   - Provides unrestricted access for OS operations like memory management and hardware control.

Importance: This separation enhances stability by isolating user processes from critical system tasks, preventing system crashes due to faulty user programs.

Q23. Describe the evolution of Operating Systems from batch systems to time-sharing systems.

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1. Batch Systems:
   - Jobs are collected, grouped, and processed sequentially.
   - No user interaction during execution.
2. Time-Sharing Systems:
   - Introduced multitasking and real-time user interaction.
   - Used time slices to share CPU among multiple users, improving efficiency.

Advancements:
Time-sharing systems enabled concurrent use of resources, making computers more interactive and accessible.

Q24. Explain how time-sharing systems enable multiple users to interact with a computer simultaneously.

- Techniques Used:
  - CPU Scheduling: Distributes CPU time in time slices to ensure responsiveness.
  - Process Isolation: Prevents interference between users' processes.
  - Memory Management: Allocates memory dynamically to active processes.

Result: These systems ensure users experience quick responses and seamless interaction, even when multiple users are active.

Q25. What are the key characteristics of a real-time operating system (RTOS)? How does it differ from general-purpose operating systems?

1. Characteristics of RTOS:
   - Deterministic: Ensures predictable response times.
   - Priority Scheduling: Executes high-priority tasks first.
   - Low Latency: Minimal delay in task execution.
2. Differences:
   - RTOS focuses on time-bound tasks (e.g., medical devices).
   - General-purpose OS prioritizes multitasking and user convenience (e.g., Windows).

Q26. Describe how a multiprocessor system works and the challenges associated with designing an OS for such a system.

1. Working of Multiprocessor Systems:
    - Multiple CPUs collaborate to execute tasks in parallel.
    - Processes are distributed across processors to improve performance.
2. Challenges:
    - Resource Sharing: Managing shared resources like memory without conflicts.
    - Load Balancing: Ensuring equal workload distribution among processors.
    - Interprocessor Communication: Synchronizing processors efficiently.

Q27. Discuss the benefits and challenges of implementing a multithreaded application. How does the Operating System support multithreading?

1. Benefits:
    - Concurrency: Executes multiple threads of a process simultaneously.
    - Resource Efficiency: Shares resources like memory among threads, reducing overhead.
    - Responsiveness: Enhances user experience by performing background tasks.
2. Challenges:
    - Synchronization Issues: Threads may access shared data inconsistently.
    - Debugging Complexity: Errors are harder to identify in concurrent environments.
3. OS Support:
    - Provides APIs for thread creation and synchronization (e.g., mutexes, semaphores).
    - Allocates CPU time to threads using scheduling algorithms.

Q28. Explain the layered structure of an Operating System. How does it help in the design and maintenance of the OS?

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1. Definition: The OS is organized into layers, each performing specific functions, from hardware interaction to user interfaces.
2. Advantages:
   - Modularity: Each layer can be developed and tested independently.
   - Ease of Maintenance: Errors are isolated within specific layers.
3. Examples: Device drivers operate at the lower layers, while user applications interact with the higher layers.

Q29. Compare and contrast monolithic kernels and microkernels in terms of structure, performance, and reliability.

1. Monolithic Kernels:
   - All OS services run in kernel mode.
   - Advantages: High performance due to direct communication.
   - Disadvantages: Lack of modularity and potential instability.
2. Microkernels:
   - Minimal functionality in kernel mode; other services run in user mode.
   - Advantages: Improved security and modularity.
   - Disadvantages: Performance overhead due to frequent mode switches.

Q30. Discuss the potential advantages of microkernel systems over monolithic systems in terms of security and modularity.

1. Security:
   - Microkernels isolate OS services in user mode, reducing the impact of system crashes or vulnerabilities.
2. Modularity:
   - Each service operates independently, allowing easier updates and debugging.

Example: If a driver fails in a microkernel system, it does not crash the entire OS, unlike monolithic kernels.

Level C: Difficult Questions (10 Marks Each)

Q31. Critically analyze the role of an Operating System in virtual memory management.

1. Role:
   - Creates an illusion of more memory than physically available by using disk storage.
   - Ensures efficient memory allocation among processes.
2. Key Functions:
   - Paging: Divides memory into fixed-size pages, mapping them to physical memory.
   - Segmentation: Divides memory into variable-sized segments for better organization.
   - Swapping: Moves inactive pages to disk to free up RAM.
3. Benefits:
   - Allows multitasking and running larger applications.
   - Improves system stability by isolating processes.

Q32. Describe process management, memory management, storage management, and mass storage management.

1. Process Management: Handles process creation, execution, and termination. Includes scheduling, synchronization, and inter-process communication.
2. Memory Management: Allocates, deallocates, and monitors memory usage to optimize performance.
3. Storage Management: Organizes and retrieves data on storage devices through file systems.
4. Mass Storage Management: Manages long-term storage devices like HDDs and SSDs for data persistence.

Q33. Evaluate the transition from batch processing systems to interactive and time-sharing systems.

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2. Interactive Systems: Introduced real-time user interaction, enhancing usability.
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Impact: This evolution increased efficiency, user convenience, and accessibility of computing systems.

Q34. Examine the architectural differences between symmetric and asymmetric multiprocessing.

1. Symmetric Multiprocessing (SMP):
   - All processors have equal access to resources and execute tasks independently.
   - Advantages: Better load balancing and resource utilization.
2. Asymmetric Multiprocessing (AMP):
   - One processor manages the system, while others perform tasks.
   - Advantages: Simplifies design but limits scalability.

Q35. Define different types of storage structure and its hierarchy.

1. Primary Storage: RAM and cache for fast, temporary data access.
2. Secondary Storage: HDDs and SSDs for long-term storage.
3. Tertiary Storage: Backup systems like optical disks and tapes.
4. Hierarchy: Organized based on speed and cost, with primary storage being fastest and most expensive.

Q36. What are operating system services and system components?

1. OS Services:
   - Process Management: Handles processes and multitasking.
   - Memory Management: Allocates memory to processes.

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   - Process Management: Handles processes and multitasking.
   - Memory Management: Allocates memory to processes.
   - File System Management: Organizes and accesses files.
2. System Components:
   - Kernel: Core of the OS managing resources.
   - Device Drivers: Facilitate hardware communication.
   - User Interface: Provides interaction with the OS (e.g., GUI).